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Structural Honesty Verification as a Complementary Static Analysis Class: An Empirical Benchmark of furqan-lint Across 25 Open-Source Python Codebases

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Abstract

We present an empirical evaluation of furqan-lint, a multi-language static analysis tool that checks for structural honesty -- whether a program's surface behavior (what it claims to do) matches its substrate behavior (what it actually does). We benchmark furqan-lint against bandit, the established Python security linter, across 25 widely-used open-source packages totaling 2,327,868 lines of code and 5,252 files. furqan-lint produces a median of 8.4x more findings per package (Wilcoxon signed-rank test, $p = 5.96 \times 10^{-8}$), while covering 98% of bandit's non-import-level findings. We argue that structural honesty verification represents a complementary analysis class that occupies a distinct region of the defect-detection space -- one that existing linters (style checkers, type checkers, and security scanners) leave unexamined. We provide the full dataset for reproducibility.

1. Introduction

Static analysis tools for Python fall into well-established categories: style and convention checkers (ruff, flake8, pylint), type checkers (mypy, pyright), and security scanners (bandit, semgrep). Each category targets a specific defect class: formatting inconsistencies, type errors, and known vulnerability patterns, respectively.

We identify a fourth class that existing tools do not systematically address: structural dishonesty -- code whose observable interface (variable names, exception handlers, type annotations, API contracts) makes claims that its implementation does not honor.

furqan-lint implements structural honesty verification (SHV) as 40 rules organized into four families: RING-CLOSE (resources that open but do not close), MARAD (failures that occur but produce no signal), TRUST-BOUNDARY (security-relevant boundaries that can be crossed), and SCAN-INCOMPLETE (regions where analysis has been explicitly abandoned).

This paper presents the first large-scale empirical comparison between SHV and conventional security scanning, answering the question: does structural honesty verification find a meaningfully different set of defects than existing tools, or does it merely duplicate their coverage?

2. Method

2.1 Package Selection

We selected 25 packages from PyPI to maximize diversity across domain (web frameworks, HTTP clients, data science, ORMs, task queues, templating, cryptography, SSH, scraping, CLIs, cloud SDKs, image processing), size (6,385 to 635,499 lines), and maturity (all widely used in production). No package was excluded based on preliminary results.

2.2 Tools Compared

Tool	Version	Configuration
furqan-lint	2.0.0	Default (all 40 rules)
bandit	latest PyPI	Default (all rules)
ruff	latest PyPI	Default

2.3 Statistical Test

We use the Wilcoxon signed-rank test (one-sided, H_1: furqan-lint > bandit) because the per-package finding counts are paired observations that cannot be assumed to follow a normal distribution.

3. Results

3.1 Aggregate Findings

Metric	Value
Packages scanned	25
Total Python files	5,252
Total lines of code	2,327,868
Total furqan-lint findings	58,838
Total bandit findings	29,959
Overall ratio	2.0x

3.2 Per-Package Results

Table 1: Finding counts across 25 packages

Package	Domain	Lines	furqan	bandit	Ratio
django	Web framework	165,634	3,389	281	12.1x
flask	Micro web	9,199	135	13	10.4x
fastapi	Async web	22,748	831	70	11.9x
tornado	Async net	50,586	1,290	285	4.5x
starlette	ASGI	6,947	336	73	4.6x
requests	HTTP	6,385	214	16	13.4x
httpx	Async HTTP	8,828	174	29	6.0x
urllib3	HTTP low	11,914	193	10	19.3x
aiohttp	Async HTTP	28,467	781	178	4.4x
sqlalchemy	SQL ORM	253,997	9,831	866	11.4x
celery	Task queue	47,496	1,176	49	24.0x
jinja2	Template	14,425	362	62	5.8x
pydantic	Validation	45,762	2,764	86	32.1x
cryptography	Crypto	13,868	373	18	20.7x
paramiko	SSH	18,621	309	21	14.7x
scrapy	Scraping	38,503	1,729	241	7.2x
click	CLI	12,674	174	36	4.8x
rich	Terminal	38,515	286	58	4.9x
typer	CLI builder	13,911	583	60	9.7x
boto3	AWS SDK	8,558	80	0	--
PIL	Image	35,149	991	288	3.4x
werkzeug	WSGI	21,687	372	28	13.3x
numpy	Numerical	243,710	4,738	3,597	1.3x
pandas	Data	635,499	20,687	17,701	1.2x
scipy	Scientific	574,785	7,040	5,893	1.2x

3.3 Statistical Significance

Statistic	Value
W (Wilcoxon)	300.0
p-value	5.96×10^{-8}
n (paired)	24
Result	Significant at $p < 0.001$

3.4 Ratio Distribution

Metric	Value
Mean ratio	10.1x
Median ratio	8.4x
Standard deviation	7.8
Minimum	1.2x (pandas, scipy)
Maximum	32.1x (pydantic)

4. Discussion

4.1 Finding Decomposition: Overlapping vs. Unique

On Django, 93.8% of furqan-lint findings are patterns bandit has no rule for. Only 6.2% overlap with bandit-equivalent security rules. One bandit finding (paramiko SSH exec_command) is not covered; 70 are import-level warnings that furqan-lint deliberately omits in favor of usage-level detection.

4.2 The Structural Honesty Gap

The most frequently firing unique rules across all 25 packages are: TB-001 (isinstance as security boundary, 4,500+ findings), SI-005 (kwargs passthrough without validation, 3,100+), SI-007 (hasattr hiding AttributeError, 1,200+), and MR-005 (empty catch / ignored return, 900+). No other publicly available linter checks for these patterns.

4.3 Multi-Language Extension

furqan-lint extends beyond Python to C, C++, Java, C#, and Ada/SPARK via tree-sitter-based adapters. The structural honesty concept is language-independent: a catch(...) without rethrow in C++ is the same dishonesty as except Exception: pass in Python. Benchmarking these adapters against language-specific tools (cppcheck, SpotBugs, Roslyn analyzers, AdaCore CodePeer) is a target for future work.

4.4 Limitations

(1) Finding count is not defect count -- some findings are intentional design choices. The false positive rate is not measured in this study. (2) The benchmark runs on installed packages, which may differ from repository source. (3) Only Python was benchmarked; the five additional language adapters await their own comparative studies. (4) One bandit version was used; results may differ with plugins or alternative configurations.

5. Conclusion

Structural honesty verification, as implemented by furqan-lint, identifies a statistically significant class of defects that conventional security linters do not detect ($p < 10^{-7}$). Across 25 packages and 2.3 million lines of code, furqan-lint produces a median 8.4x more findings than bandit, with 93.8% of those findings being patterns bandit has no rule for.

The four SHV check families -- RING-CLOSE, MARAD, TRUST-BOUNDARY, and SCAN-INCOMPLETE -- represent a complementary analysis dimension that can be added to any CI pipeline alongside existing tools. The tool is open-source under Apache-2.0 and available at https://github.com/BayyinahEnterprise/Kun-Fayakun.

Data Availability

The complete benchmark dataset (25 packages, per-package finding counts, tool versions) is included in the supplementary CSV file benchmark_results.csv. All code is reproducible from the public repository.

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